BASES DE DATOS

UD6

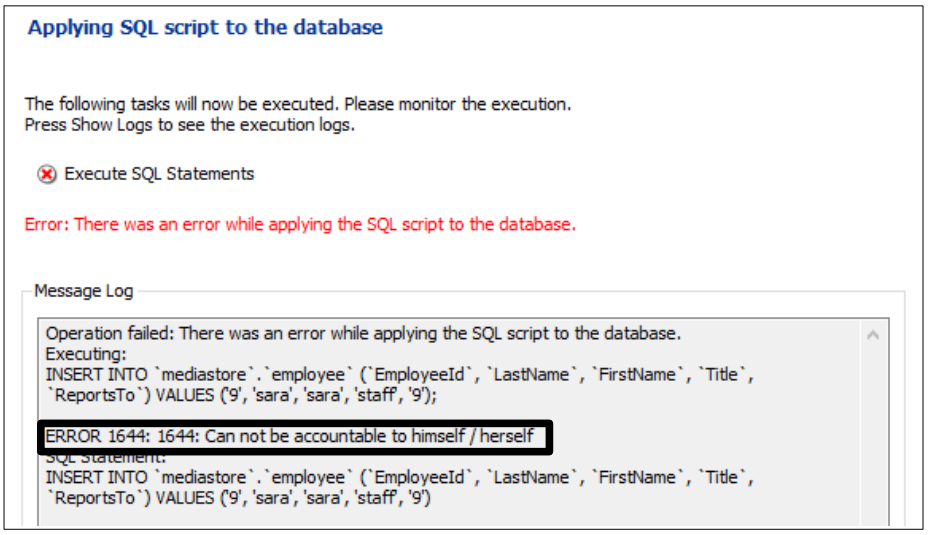
NEREA MONTOYA OLIVARES

DRIVE Link: https://drive.google.com/file/d/1-jchpcPDvjOQ7Qzvdw4fXNZ23qOI881x/view?usp=drive_link

Exercise 1

a) An employee can not be accountable to himself / herself. So we need to create a trigger that checks before inserting into employee if an employee reports to himself / herself. If it is, it should throw an error indicating if the failure with a message. Show how you check the trigger.

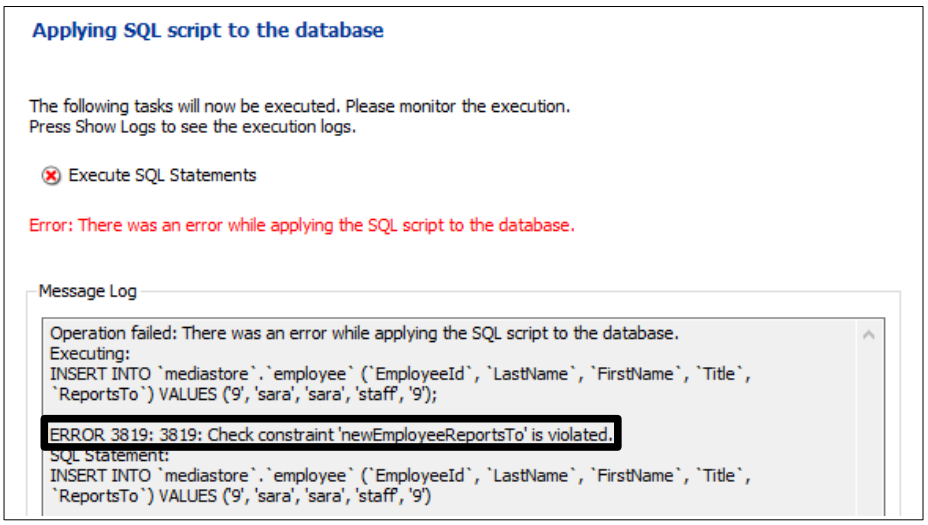
```
DELIMITER //
CREATE TRIGGER insertNewEmployee BEFORE INSERT ON employee
FOR EACH ROW
BEGIN
        IF NEW.EmployeeId = NEW.ReportsTo THEN
                SIGNAL SQLSTATE '45000'
                SET MESSAGE_TEXT = 'Can not be accountable to himself / herself';
        END IF;
END//
DELIMITER ;
```



b) Can you implement this restriction without a trigger? How?

Yes, using a CHECK constraint.

```
ALTER TABLE employee
ADD CONSTRAINT newEmployeeReportsTo
CHECK (reportsTo <> employeeId);
```



Exercise 2

We want to keep track every time a customer changes his/her email. For this reason, we will create a table to save the previous emails. The table has the following structure:

customerId: int,
time_stamp: datetime,
email: varchar(60),

At the time_stamp field we will register the current time (when the email is changed). Use as primary key the combination of customerId and time_stamp.

Create a trigger (every time an update is done on Customer table) to fill this table. Take into account that it doesn't matter if other customer data are changed, we only need to know when the 'email' field is changed.Show how you check the trigger.

```
USE mediastore;
CREATE TABLE oldEmail
(
customerId INT,
time_stamp DATETIME DEFAULT CURRENT_TIMESTAMP,
email VARCHAR(60),
PRIMARY KEY (customerId, time_stamp)
);

DELIMITER //
CREATE TRIGGER before_update_costumer
AFTER UPDATE ON customer
FOR EACH ROW
BEGIN
        IF NEW.email <> OLD.email THEN
        INSERT INTO oldEmail (customerId, time_stamp, email)
        VALUES (NEW.customerId, NOW(), OLD.email);
        END IF;
END//
DELIMITER ;
```

✓	1	21:32:17	USE mediastore	0 row(s) affected
✓	2	21:32:17	CREATE TABLE oldEmail (customerId INT, time_stamp DATETIME DEFAULT CURRENT_TIMESTA...	0 row(s) affected
✓	3	21:32:17	CREATE TRIGGER before_update_costumer AFTER UPDATE ON customer FOR EACH ROW BEG...	0 row(s) affected

---

BEFORE:

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

CustomerId	FirstName	LastName	Company	Address	City	State	Country	PostalCode	Phone	Fax	Email
1	Luís	Gonçalves	Embraer - E...	Av. Brigadeiro Fari...	São Jo...	SP	Brazil	12227-000	+55 (12) 3923...	+55 ...	luisg@embraer.com.br

AFTER:

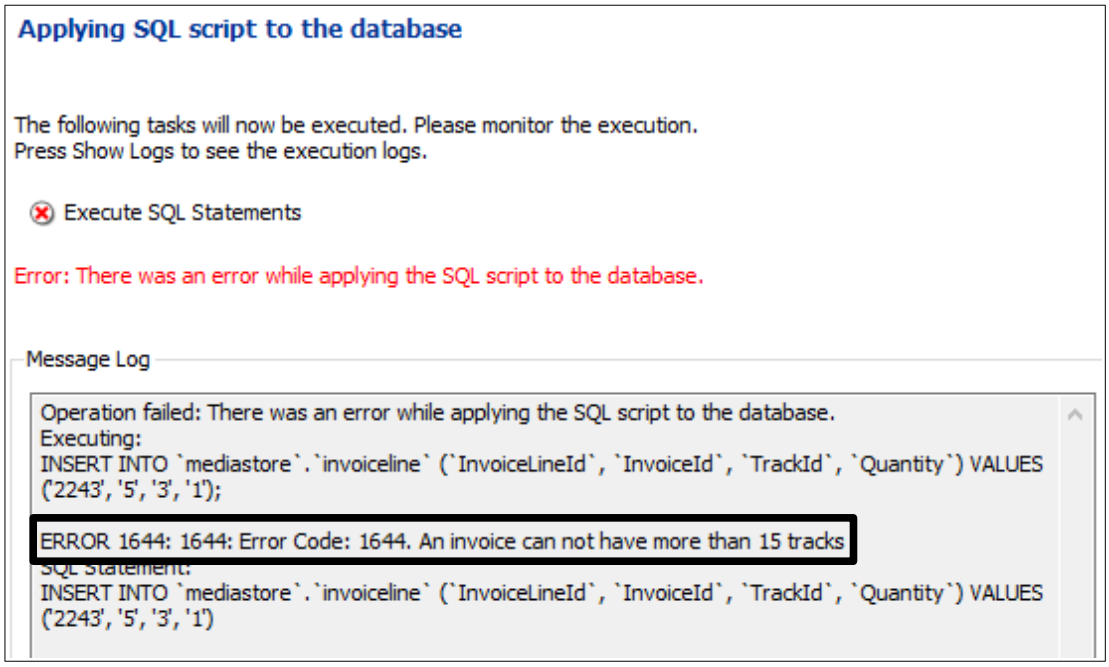
Result Grid | Filter Rows: | Edit:

customerId	time_stamp	email
1	2024-03-07 21:35:01	luisg@embraer.com.br

Exercise 3

An invoice can not have more than 15 lines or tracks. Throw an error indicating the failure with a message. Show how you check the trigger.

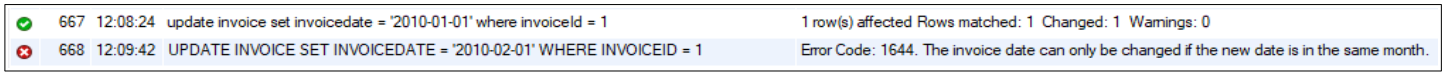
```
DELIMITER //
CREATE TRIGGER before_insert_invoiceline BEFORE INSERT ON
invoiceline FOR EACH ROW
BEGIN
    DECLARE rowcount INT;
    SELECT COUNT(*) INTO rowcount FROM invoiceline WHERE InvoiceId=NEW.InvoiceId;
    IF rowcount > 15 THEN
        SIGNAL SQLSTATE'45000'
        SET MESSAGE_TEXT='Error Code: 1644. An invoice can not have more than 15 tracks';
    END IF;
END//
DELIMITER ;
```



Exercise 4

The invoice date can only be changed if the new date is in the same month. Create a trigger to check that restriction. Show how you check the trigger.

```
DELIMITER //
CREATE TRIGGER before_update_invoice
AFTER UPDATE ON invoice
FOR EACH ROW
BEGIN
        DECLARE old_Month INT;
        DECLARE new_Month INT;
        SET old_Month = month(old.InvoiceDate);
        SET new_Month = month(new.InvoiceDate);
        IF new_Month <> old_Month THEN
                SIGNAL SQLSTATE'45000'
                SET MESSAGE_TEXT='The invoice date can only be changed if the new date is in
                the same month.';
        END IF;
END //
DELIMITER ;
```



Exercise 5

Create a procedure to remove an employee. For avoiding to get an error about foreign key constraints fails, pay attention to:

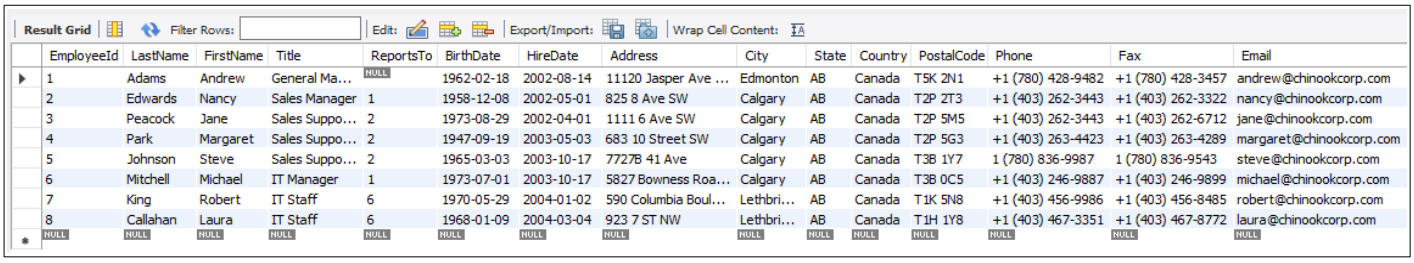
- **employees reporting to the employee that will be removed. Now they must report to the employee boss that will be removed.**
- **customers that are supported by the employee that will be removed. Now they must be supported by the employee boss that will be removed.**

Use an employee identifier as parameter. Customize error for any employee identifier that doesn't exist.

```
DELIMITER //
CREATE PROCEDURE delete_employee (IN employee_id INT)
BEGIN
	DECLARE boss_id INT;
	SELECT ReportsTo INTO boss_id FROM employee WHERE EmployeeId = employee_id;
	IF boss_id IS NULL THEN
		SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT = 'Employee does not exist';
	END IF;
	UPDATE employee SET ReportsTo = boss_id WHERE ReportsTo = employee_id;
	UPDATE customer SET SupportRepId = (SELECT ReportsTo FROM employee WHERE
	employeeId = employee_id) WHERE supportRepId = employee_id;
	DELETE FROM employee WHERE EmployeeId = employee_id;
END//
DELIMITER ;

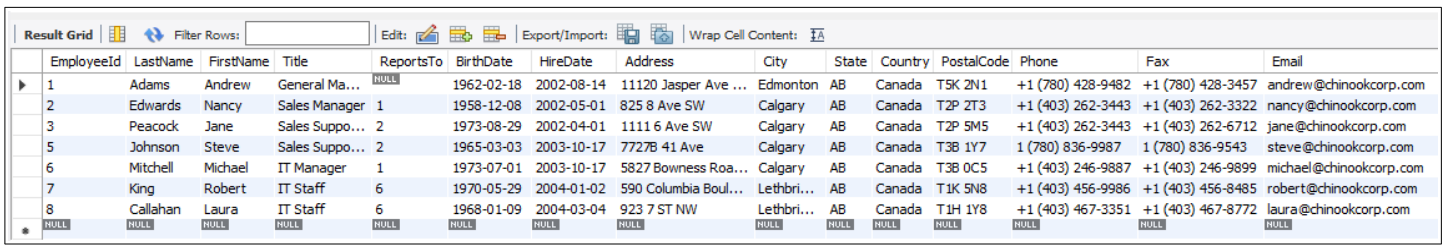
CALL delete_employee (4);
```

BEFORE:



EmployeeId	LastName	FirstName	Title	ReportsTo	BirthDate	HireDate	Address	City	State	Country	PostalCode	Phone	Fax	Email
1	Adams	Andrew	General Ma...	NULL	1962-02-18	2002-08-14	11120 Jasper Ave ...	Edmonton	AB	Canada	T5K 2N1	+1 (780) 428-9482	+1 (780) 428-3457	andrew@chinookcorp.com
2	Edwards	Nancy	Sales Manager	1	1958-12-08	2002-05-01	825 8 Ave SW	Calgary	AB	Canada	T2P 2T3	+1 (403) 262-3443	+1 (403) 262-3322	nancy@chinookcorp.com
3	Peacock	Jane	Sales Suppo...	2	1973-08-29	2002-04-01	1111 6 Ave SW	Calgary	AB	Canada	T2P 5M5	+1 (403) 262-3443	+1 (403) 262-6712	jane@chinookcorp.com
4	Park	Margaret	Sales Suppo...	2	1947-09-19	2003-05-03	683 10 Street SW	Calgary	AB	Canada	T2P 5G3	+1 (403) 263-4423	+1 (403) 263-4289	margaret@chinookcorp.com
5	Johnson	Steve	Sales Suppo...	2	1965-03-03	2003-10-17	7727B 41 Ave	Calgary	AB	Canada	T3B 1Y7	1 (780) 836-9987	1 (780) 836-9543	steve@chinookcorp.com
6	Mitchell	Michael	IT Manager	1	1973-07-01	2003-10-17	5827 Bowness Roa...	Calgary	AB	Canada	T3B 0C5	+1 (403) 246-9887	+1 (403) 246-9899	michael@chinookcorp.com
7	King	Robert	IT Staff	6	1970-05-29	2004-01-02	590 Columbia Boul...	Lethbri...	AB	Canada	T1K 5N8	+1 (403) 456-9986	+1 (403) 456-8485	robert@chinookcorp.com
8	Callahan	Laura	IT Staff	6	1968-01-09	2004-03-04	923 7 ST NW	Lethbri...	AB	Canada	T1H 1Y8	+1 (403) 467-3351	+1 (403) 467-8772	laura@chinookcorp.com

AFTER:



EmployeeId	LastName	FirstName	Title	ReportsTo	BirthDate	HireDate	Address	City	State	Country	PostalCode	Phone	Fax	Email
1	Adams	Andrew	General Ma...	NULL	1962-02-18	2002-08-14	11120 Jasper Ave ...	Edmonton	AB	Canada	T5K 2N1	+1 (780) 428-9482	+1 (780) 428-3457	andrew@chinookcorp.com
2	Edwards	Nancy	Sales Manager	1	1958-12-08	2002-05-01	825 8 Ave SW	Calgary	AB	Canada	T2P 2T3	+1 (403) 262-3443	+1 (403) 262-3322	nancy@chinookcorp.com
3	Peacock	Jane	Sales Suppo...	2	1973-08-29	2002-04-01	1111 6 Ave SW	Calgary	AB	Canada	T2P 5M5	+1 (403) 262-3443	+1 (403) 262-6712	jane@chinookcorp.com
5	Johnson	Steve	Sales Suppo...	2	1965-03-03	2003-10-17	7727B 41 Ave	Calgary	AB	Canada	T3B 1Y7	1 (780) 836-9987	1 (780) 836-9543	steve@chinookcorp.com
6	Mitchell	Michael	IT Manager	1	1973-07-01	2003-10-17	5827 Bowness Roa...	Calgary	AB	Canada	T3B 0C5	+1 (403) 246-9887	+1 (403) 246-9899	michael@chinookcorp.com
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Exercise 6

Create a procedure with three parameters. The first one represents the title of an album. The procedure must find the artist name of the album and return it through an output parameter (the second one).
Also, the procedure must find all the tracks with their duration between brackets (format mm:ss) from this album, and return the result through a new output parameter (the third one).

```
DELIMITER //
CREATE PROCEDURE track_from_album (
IN album_title VARCHAR(255),
OUT artist_name VARCHAR(255),
OUT track_list VARCHAR(1000)
)
BEGIN
	DECLARE album_id INT;

	SELECT AlbumId INTO album_id FROM album WHERE Title = album_title;

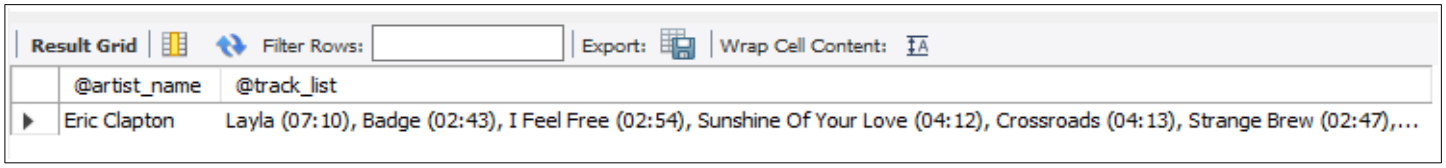
	IF album_id IS NULL THEN
		SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT = 'Album does not exist';
	END IF;

	SELECT NAME INTO artist_name
	FROM album
	JOIN artist ON album.ArtistId = artist.ArtistId
	WHERE album.AlbumId = album_id;

	SELECT GROUP_CONCAT(CONCAT(Name, ' (', LPAD(FLOOR(Milliseconds DIV 60000), 2,
	'0'), ':', LPAD(TIME_FORMAT((Milliseconds DIV 1000) % 60, '%s'), 2, '0'), ')') ORDER BY
	TrackId SEPARATOR ', ') INTO track_list
	FROM track
	WHERE AlbumId = album_id;
END//
DELIMITER ;

CALL track_from_album('The Cream Of Clapton',@artist_name,@track_list);

SELECT @artist_name, @track_list;
```



Exercise 7

Create a function that takes a track id as a parameter and shows information about the number of playlists and invoices where is found that track. This information must be shown as a string.

```
DELIMITER //

CREATE FUNCTION GetTrackInfo(track_id INT) RETURNS VARCHAR(1000)
DETERMINISTIC
BEGIN
	DECLARE playlists_count INT;
	DECLARE invoices_count INT;
	DECLARE info VARCHAR(1000);

	SELECT COUNT(*) INTO playlists_count
	FROM PlaylistTrack
	WHERE TrackId = track_id;

	SELECT COUNT(*) INTO invoices_count
	FROM InvoiceLine
	WHERE TrackId = track_id;

	SET info = CONCAT('Number of playlists: ', playlists_count, ', Number of invoices: ',
	invoices_count);

	RETURN info;
END//

DELIMITER ;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

TrackId	Name	Composer	foundIn
1	For Those About T...	Angus Young, M...	Number of playlists: 3, Number of invoices: 1
2	Balls to the Wall	NULL	Number of playlists: 3, Number of invoices: 2
3	Fast As a Shark	F. Baltes, S. Kau...	Number of playlists: 4, Number of invoices: 1
4	Restless and Wild	F. Baltes, R.A. S...	Number of playlists: 4, Number of invoices: 1
5	Princess of the Dawn	Deaffy & R.A. S...	Number of playlists: 4, Number of invoices: 1
6	Put The Finger On ...	Angus Young, M...	Number of playlists: 2, Number of invoices: 1
7	Let's Get It Up	Angus Young, M...	Number of playlists: 2, Number of invoices: 0
8	Inject The Venom	Angus Young, M...	Number of playlists: 2, Number of invoices: 2
9	Snowballed	Angus Young, M...	Number of playlists: 2, Number of invoices: 2
10	Evil Walks	Angus Young, M...	Number of playlists: 2, Number of invoices: 1
11	C.O.D.	Angus Young, M...	Number of playlists: 2, Number of invoices: 0
12	Breaking The Rules	Angus Young, M...	Number of playlists: 2, Number of invoices: 1